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PHHP Days 2024

Effect of early and adequate empiric antibiotic therapy for hospital-acquired sepsis in the age of antimicrobial resistance: a target trial emulation using machine learning

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Mentors: Dr. Mattia Prosperi & Dr. J Glenn Morris

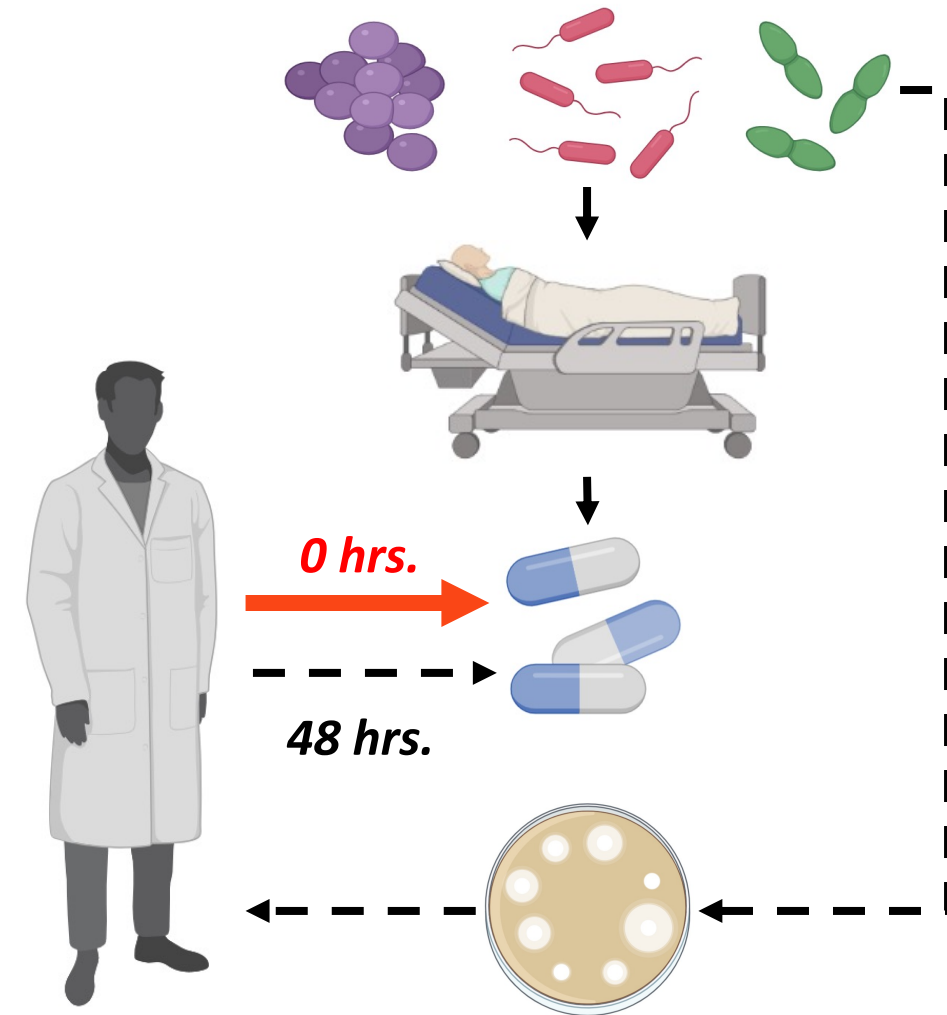
COLLEGE *of* **PUBLIC HEALTH & HEALTH PROFESSIONS**

BACKGROUND & SIGNIFICANCE

- Hospital-acquired infections are major health safety issues worldwide
- Severe cases can ultimately progress into sepsis → high morbidity/mortality
- At the time of diagnosis, we do not know the exact pathogen... What do we do?
 - Empiric antimicrobial therapy, but imperfect
 - Guidelines exist, but with a lot of nuance
 - Antimicrobial resistance complicates further

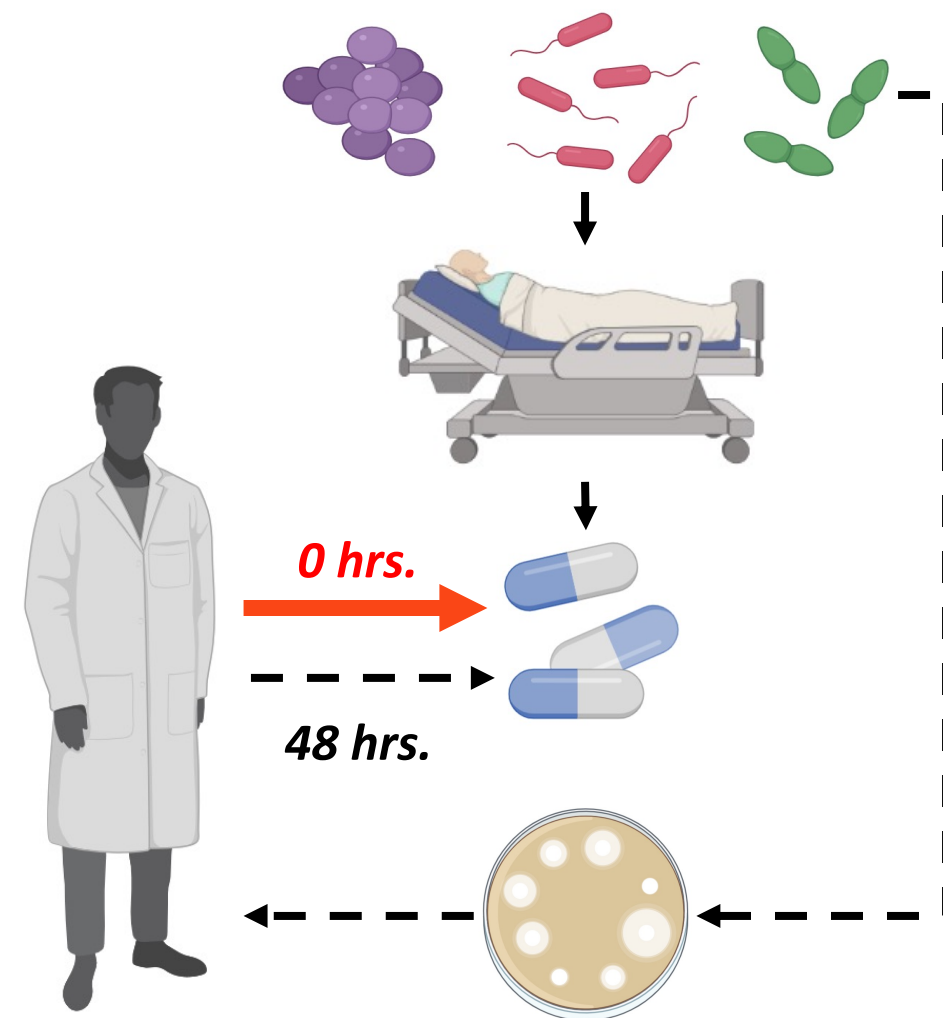
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What is the causal effect of prescribing adequate therapy on mortality? *Average Treatment Effect (ATE)*

Who may most benefit from “broader” empiric therapy treatments? *Conditional ATE*

METHODS: Target Trial Emulation + Machine Learning

What is the causal effect of prescribing adequate antimicrobial therapy within 24 hours on mortality in hospital-onset sepsis?

- **Randomized controlled trials** are the 'gold standard' for answering causal questions
- RCTs are not always feasible, ethical, time-constraint...etc.
- Is the RCT approach appropriate for this research question? No...
- Observational based on real-world data is required with limitations (50+ biases)
- **Solution:** Target trial emulation = observational study that emulates the design of RCT

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Eligibility

Hospital-Onset
Sepsis

Treatment

1. *Adequate Abx*
2. *Inadequate Abx*

Assignment

*Randomization
emulated baseline +
time-varying
covariates with ML**

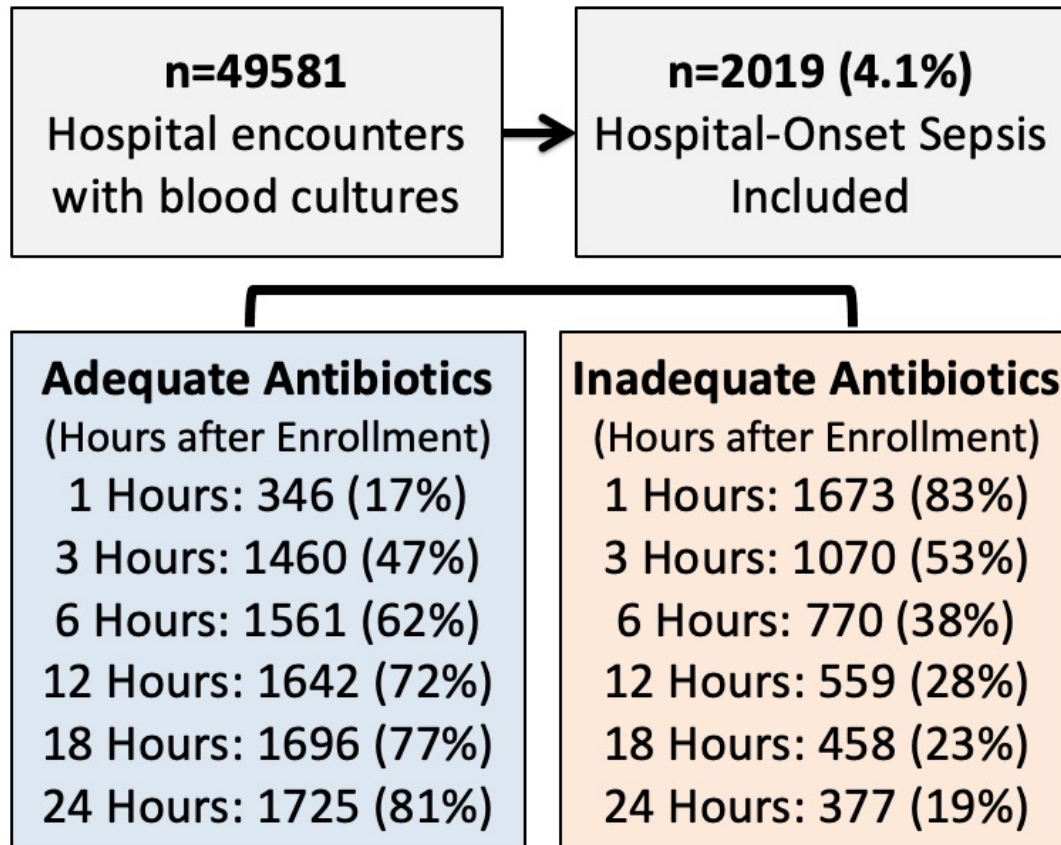
Outcome

*30-day in-hospital
mortality*

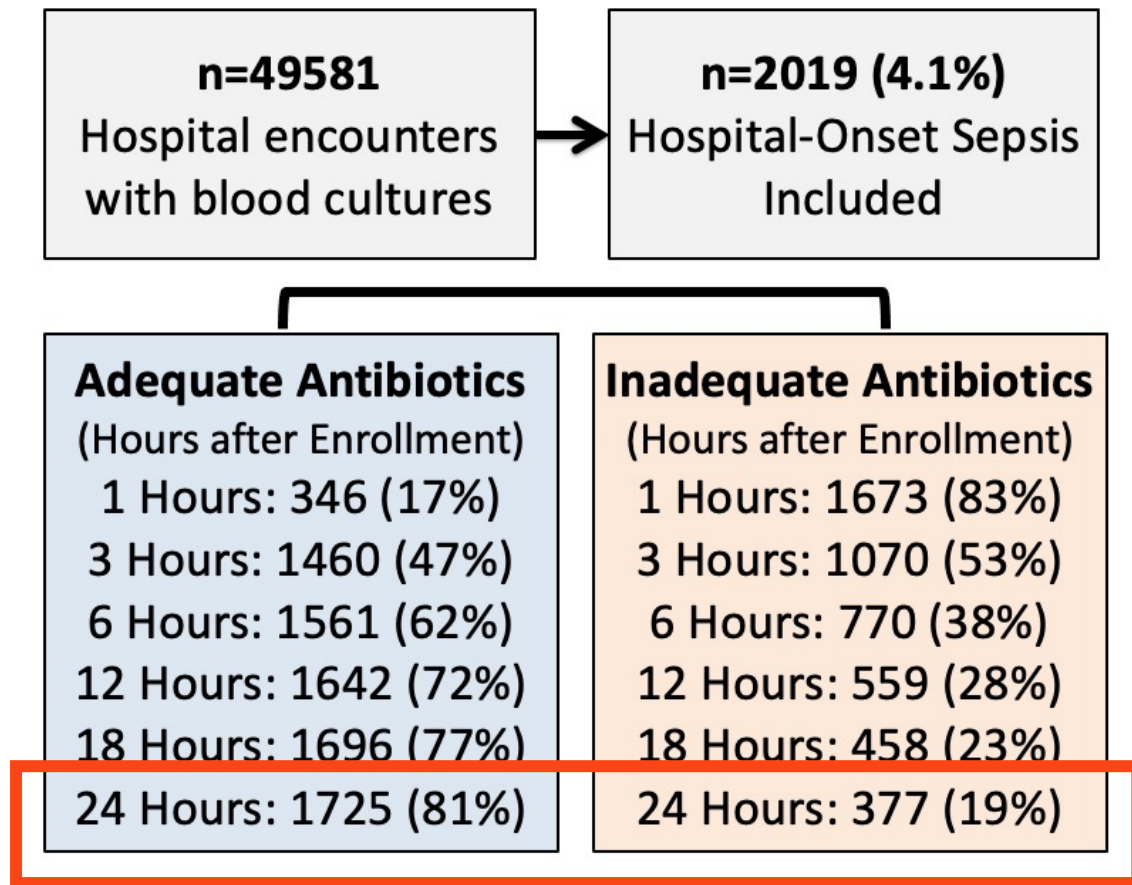
Casual Contrast

*Analog of Per-
protocol effect*

RESULTS: Population Summary – UF IDR (~140,000 Encounters)



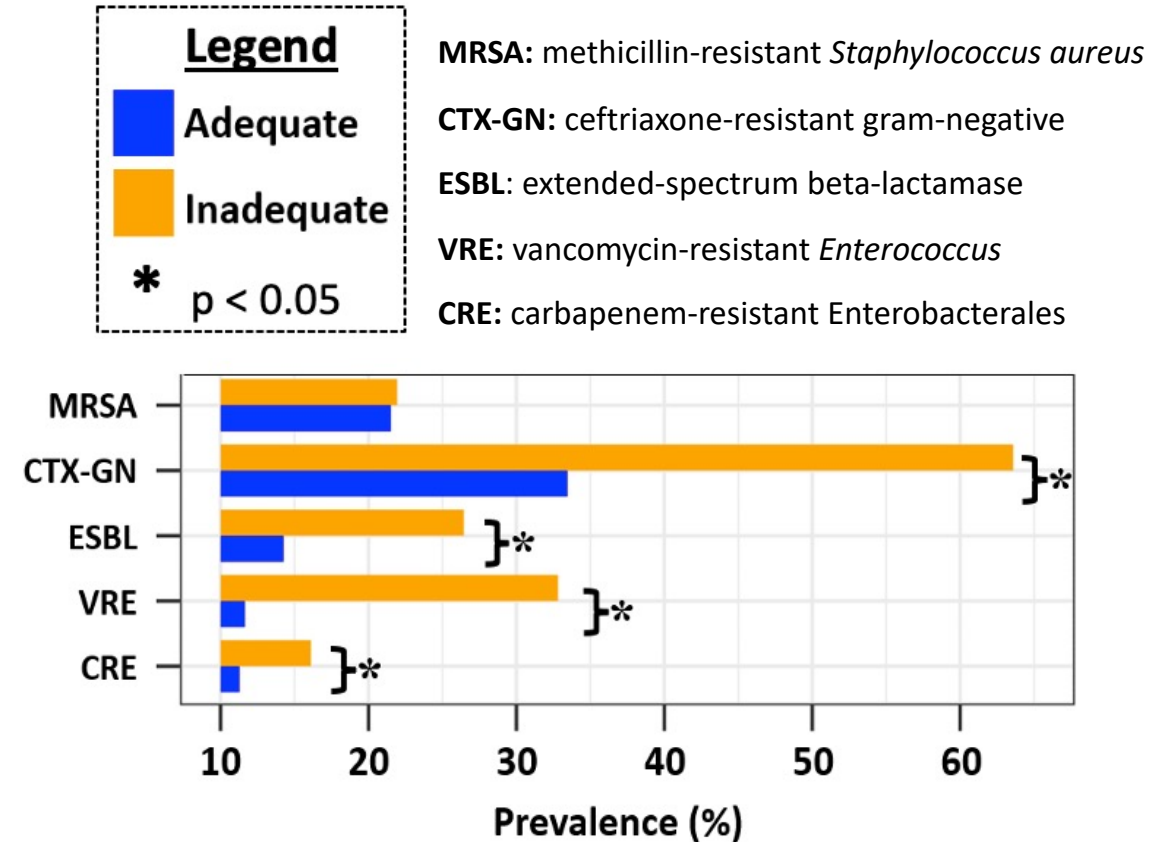
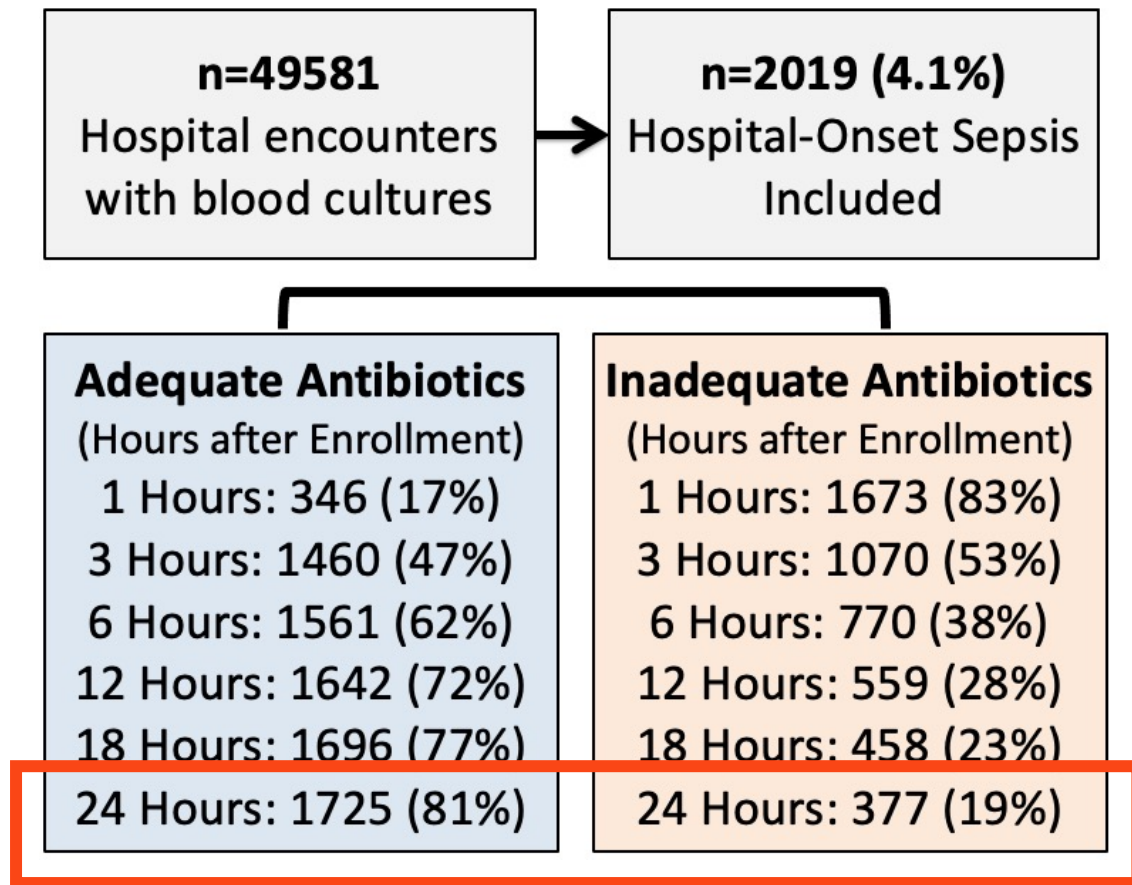
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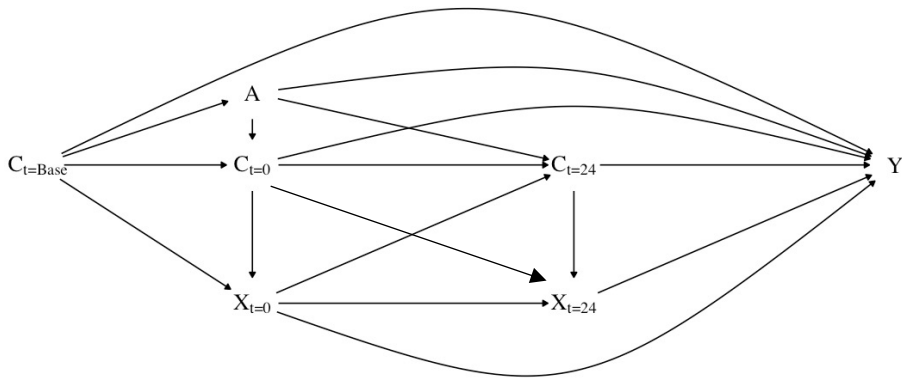
↑
1 in 5

RECEIVED INADEQUATE ANTIBIOTIC THERAPY

RESULTS: Population Summary – UF IDR (~140,000 Encounters)



RESULTS: Causal Modelling – Marginal Structural Cox Hazards



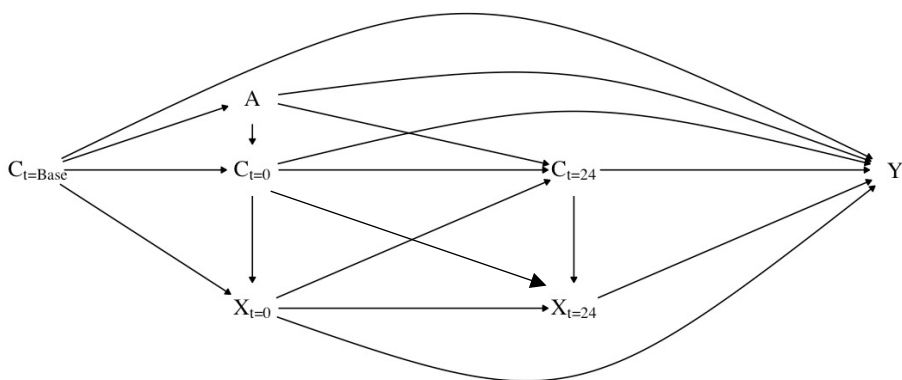
“Randomization”

Inverse-Probability of
Treatments Weights Testing
Machine Learning Approaches



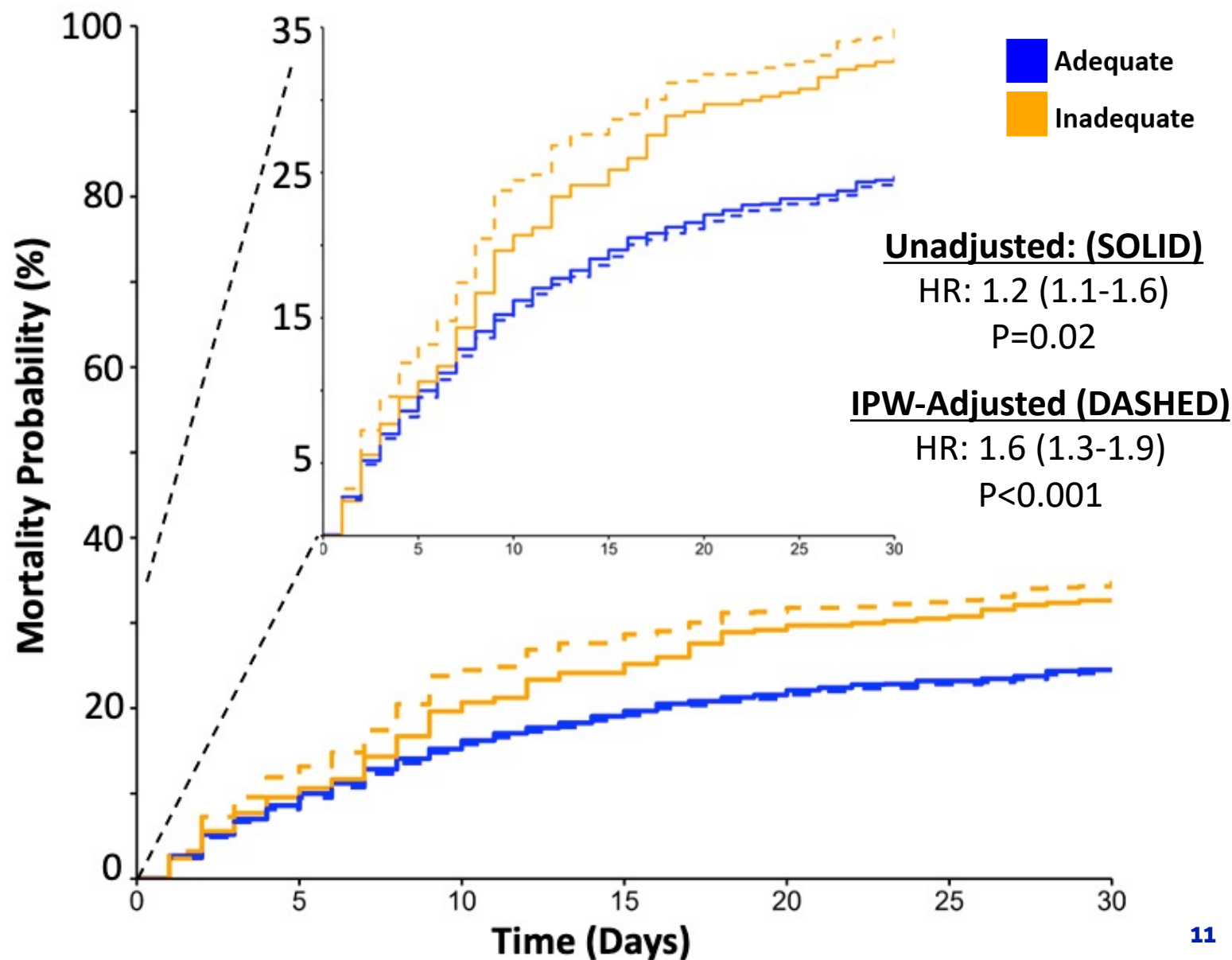
Fit Marginal Structural Cox
Proportional Hazards Model

RESULTS: Causal Modelling – Marginal Structural Cox Hazards



↓
“Randomization”
Inverse-Probability of
Treatments Weights at 24 Hours

↓
Fit Marginal Structural Cox
Proportional Hazards Model



CONCLUSION

1. We reinforce the critical importance of timely and adequate antimicrobial administration in hospital-onset sepsis
2. Inadequate empiric therapy was largely driven by AMR
3. We highlight the value of integrating real-world data sources, with expert-driven modeling for causal inference in complex clinical scenarios

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Observational studies are what we do when we cannot conduct a randomized trial

- We need to keep improving them -

Target trial emulation + machine learning can offer solutions



Authors & Contributors

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